

Speak Like A Mathematician:

indirect proof:

proof by contradiction:

Triangle Inequality Theorem:

Sides, Angles, and the Triangle Inequality

If one **SIDE** of a triangle is longer than another, then the **ANGLE** opposite the longer side is _____ than the angle opposite the shorter side.

If one **ANGLE** of a triangle is larger than another, then the **SIDE** opposite the larger angle is _____ than the side opposite the smaller angle.

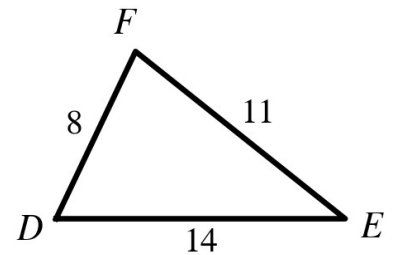
Triangle Inequality Theorem: the sum of any two side lengths is _____ than the third side length. You must check _____ sums.

Range for a third side: (difference of the two sides) < third side < (_____ of the two sides).

Example 1 - Order the Angles

In $\triangle DEF$, $DE = 14$, $EF = 11$, and $DF = 8$. Write the angles in order from smallest to largest.

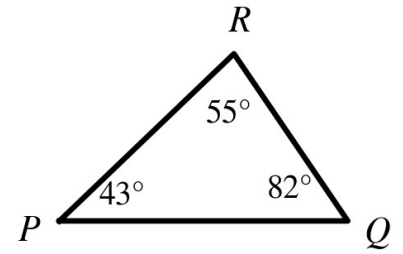
- Which angle is opposite each side?
- Order the angles.



Example 2 - Order the Sides

In $\triangle PQR$, $m\angle P = 43^\circ$, $m\angle Q = 82^\circ$, and $m\angle R = 55^\circ$. Write the sides in order from smallest to largest.

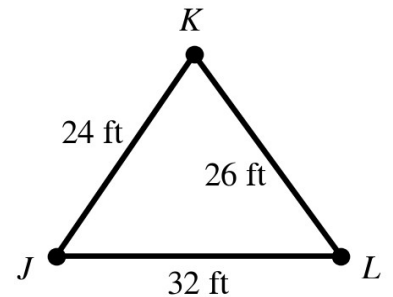
- Which side is opposite each angle?
- Order the sides.



Example 3 - A Stage Prop

A triangular mountain prop has a bottom edge of 32 ft, a left slope of 24 ft, and a right slope of 26 ft. List the angles of $\triangle JKL$ from smallest to largest.

- Name the side opposite each angle.
- Order the angles.
- Check the Triangle Inequality Theorem for this prop.



Example 4 - Can These Be the Sides of a Triangle?

Check all three sums. Explain each answer.

- a) 7, 10, 19
- b) 2.3, 3.1, 4.6
- c) $n + 6$, $n^2 - 1$, $3n$ when $n = 4$

Example 5 - More Practice

- a) 6.2, 7, 9
- b) 8, 13, 21
- c) $t - 2$, $4t$, $t^2 + 1$ when $t = 4$

Example 6 - The Range for a Third Side

Two sides of a triangle measure 8 in. and 13 in.

- a) Write the two inequalities the third side x must satisfy.
- b) Write the range as one compound inequality.

Example 7 - Writing an Indirect Proof

An indirect proof has three moves: assume the opposite, reason until something impossible happens, and conclude that the assumption was wrong.

Prove: A triangle cannot have two right angles.

Step 1) Assume the opposite of what you want to prove.

Step 2) Reason from the assumption until you reach a contradiction.

Step 3) State the conclusion.

Example 8 - Set Up an Indirect Proof

Write only the FIRST step — the assumption — for each statement.

- a) Prove: In $\triangle ABC$, if $AB \neq AC$, then $\angle C \neq \angle B$.
- b) Prove: A triangle cannot have two obtuse angles.
- c) Prove: If x^2 is odd, then x is odd.